



Particle accelerator flip

Physicists at MIT are flipping the particle accelerator model to map the structure of an atom. <http://digit.in/apr21-57>



Detecting quantum transmissions

Researchers at the Max Planck Institute of Quantum Optics develop method to track quantum transmissions. <http://digit.in/apr21-58>

Greatorex, an instrument maker and apprentice clockmaker, whose own version was too rough for any useful purpose. In a series of experiments, Boyle began to understand how gases behave under conditions of different temperatures and pressures, allowing him to gain the insight of formulating Boyle's law, which states that the pressure is inversely proportional to the volume. According to Boyle's law, as the volume increases, the pressure of the gas decreases proportionally. In fact, according to science historians, Hooke seems to have helped Boyle considerably with the mathematical equations in his studies. Boyle himself referred to the law as "Mr Townley's Hypothesis", who was a collaborator known as Richard Towneley. Towneley and Henry Power had conducted a series of experiments on gases using a barometer, and had noticed a relationship between pressure and volume, and were the first to actually uncover the working behind Boyle's Law.

Based on these findings, Denis Papin, another associate of Boyle, invented what was known as the Steam Digester. This was a device that cooked bones in a high pressure environment, allowing the extraction of fats from bones. The steam digester was the forerunner of the modern pressure cooker and the autoclave. Papin then implemented a steam release valve in case the pressure inside the vessel got too high, this

was after a number of early versions and models blew up. After a series of exploding pressure cookers, Papin finally managed to hit upon a design that did not explode. Papin noted the safety release valve, which could have a weight placed on top of it to control when the pressure would be released, would periodically rise and fall as the pressure from the steam built up and was released. Papin conceptualised the possibility of a piston that works on the pressure build up and release, but did not do anything to actually create such a device.

In 1698, Thomas Savery described a machine that he had created that used steam to generate a partial vacuum in a chamber, which in turn was used to pump up water. Savery suggested that the device could be used to drain water from pumps. While it did not have any moving parts or pistons, the device was also called a steam engine. Papin came across Savery's design, and developed a second steam engine that cannot really be called a steam engine because it worked on atmospheric pressure rather than steam. Thomas Newcomen, an ironmonger who supplied equipment to miners and looking to improve on the machinery used in the trade, developed the atmospheric engine, a device that combined the principles of both Papin's steam digester, and Savery's steam engine. The atmospheric engine was used to drain water out of mines, in 1712.



The original Magdeburg Hemispheres and Guericke's vacuum pump at Deutsches Museum in Germany.

In the 1760s, Joseph Black was investigating how various substances behave at different temperatures and pressures, and discovered one of the fundamental concepts of thermodynamics, that is latent heat, the heat that is absorbed or lost during thermodynamic processes. He came to this understanding when heating a combination of water and ice, or water and steam, and noticed that in these experiments, the proportion of water or steam increased, but the heat of the entire system did not. He shared these insights with his friend, a Scottish inventor and engineer by the name of James Watt. Watt would use this knowledge to improve on the 1712 design by Savery, and develop and improve his own version of the steam engine. Watt's biggest contribution was to have a discrete condenser for the engine, which significantly improved its efficiency. The new design proved to be wildly popular, one of the driving forces of the industrial revolution, and had steam engines pulling carriages snaking across the continent by the end of the century. Sadi Carnot, the father of Thermodynamics, wrote his treatise *Reflections on the Motive Power of Fire* based on the works of these steam engine pioneers, paving the way for later theories by Rankine and Maxwell. 



One of Watt's double acting steam engine